Khalid Hasan Raj

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Current Position

Lecturer **May 2026 - Present**
Department of Biochemistry and Molecular Biology, University of Dhaka, Dhaka, Bangladesh

Education

M.S. in Biochemistry and Molecular Biology, University of Dhaka **2025**
Thesis Title: *Hernessing E. coli for cost effective production and purification of a novel pan-serotype multi-epitope vaccine against dengue virus.*
Thesis Supervisor: Professor A.H.M. Nurun Nabi, PhD | **GPA:** 4.00/4.00 | **Rank:** 1/60

B.S. in Biochemistry and Molecular Biology, University of Dhaka **2023**
CGPA: 3.96/4.00 | **Rank**: 1/60

Work Experience

Lecturer **May 2026 - Present**
Department of Biochemistry and Molecular Biology, University of Dhaka
- Delivering Biochemistry, Bioinformatics, Molecular Biology, and Nutrition-related undergraduate courses; designing laboratory modules; preparing assessments and evaluating exam scripts; mentoring thesis students; and participating in departmental curriculum development.
- Establishing cell culture facilities and optimizing hybridoma-based monoclonal antibody production pipelines as a Research Associate.
- Doing multi-omics bioinformatics projects to develop risk prediction models for stomach and bladder cancer.
- Investigating genetic associations underlying non-communicable diseases.
- Collaborating within multidisciplinary research teams, contributing to data analysis, interpretation, and scientific writing.
- Developing research ideas and preparing research proposals and grant applications.

Research Fellow **October 2025 - May 2026**
International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b)
- Conducted mammalian cell culture and virus culture under biosafety-compliant conditions.
- Performed immunological and virological assays.
- Contributed to biosimilar potency testing and support clinical trials of viral vaccines.
- Participated in antiviral drug discovery projects.
- Collaborated in multidisciplinary research teams and contributed to data analysis, interpretation, and scientific writing.

Senior Research Assistant **June 2025 - September 2025**
International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b)
- Maintained mammalian cell lines and assisted in virus culture experiments.
- Supported laboratory operations, data collection, and experimental optimization.
- Assisted in training junior researchers and interns in laboratory techniques and safety practices.

MS Thesis Student **September 2023 - June 2025**
Department of Biochemistry and Molecular Biology, University of Dhaka
Thesis Title: *Hernessing E. coli for cost effective production and purification of a novel pan-serotype multi-epitope vaccine against dengue virus.*
- Designed an immunoinformatics-based multi-epitope vaccine targeting all dengue virus serotypes.
- Cloned the vaccine construct gene into an expression vector and expressed it in *E. coli*.
- Optimized induction conditions and purified the recombinant protein using chromatographic techniques.
- Confirmed protein expression and purity by SDS-PAGE and Western blotting.

Publications

Discovery of Quercetin as a Potential Entry Inhibitor of Nipah Virus: A Path Toward Antiviral Therapy
Alam, M.M., Rahman, M.M., **Raj, K.H.**, Nahid, A.H., Saha, P., Hossain, A., Amaya, M., Laing, E.D., Satter, S.M., Broder, C.C. and Hossain, M.E., 2026. Discovery of Quercetin as a Potential Entry Inhibitor of Nipah Virus: A Path Toward Antiviral Therapy. International Journal of Molecular Sciences, 27(18), p.8026.

Genetic association of vitamin D receptor variants and 25-hydroxyvitamin D levels with breast cancer risk and triple-negative phenotype in Bangladesh
AMowla, A., Mahmud, Z., **Raj, K. H.**, Tamanna, S., Zeba, Z., Ansari, F., & Howlader, M. Z. H. (2026). Genetic association of vitamin D receptor variants and 25-hydroxyvitamin D levels with breast cancer risk and triple-negative phenotype in Bangladesh. *Scientific Reports*. https://doi.org/10.1038/s41598-026-68260-9

A robust comprehensive immunoinformatics approach for designing a potential multi-epitope based vaccine against a reiterated monkeypox virus
Raj, K. H., Hossain, E., Zahin, H., Noman, A. A., Saba, A. A., Sayem, M., Yasmin, T., & Nabi, A. H. M. (2025). A robust comprehensive immunoinformatics approach for designing a potential multi-epitope based vaccine against a reiterated monkeypox virus. *Biochemistry and Biophysics Reports*, 43, 102075. https://doi.org/10.1016/j.bbrep.2025.102075

Comprehensive *in silico* characterization of nonsynonymous SNPs in the human ezrin (EZR) gene and their role in disease pathogenesis

Akter, S., Fuad, M., Mahmud, Z., Tamanna, S., Sayem, M., **Raj, K. H**., & Howlader, M. Z. H. (2025). Comprehensive *in silico* characterization of nonsynonymous SNPs in the human ezrin (EZR) gene and their role in disease pathogenesis. *Biochemistry and Biophysics Reports*, 42, 101972. https://doi.org/10.1016/j.bbrep.2025.101972

Skills

Molecular Biology and Biochemistry: DNA and RNA extraction, PCR, qRT-PCR, gel electrophoresis, molecular cloning and plasmid handling, heterologous protein expression, SDS-PAGE and western blotting.

Cell Culture and Virology: Mammalian cell culture, virus culture, plaque assay, pseudovirus virus neutralization assay (PNA), focus reduction neutralization test (FRNT), biosimilar potency testing and antiviral drug screening.

Immunological Techniques: Enzyme-linked immunosorbent assay (ELISA), neutralization assays and immune response testing.

Bioinformatics: Genomics and transcriptomics data analysis, immunoinformatics-based data analysis, protein structure prediction and mutation analysis, molecular docking, molecular dynamics simulation and machine learning model development.

Programming and data analysis: **R** (biostatistics, visualization, transcriptomics analysis), **Python** (data processing, machine learning algorithms implementation), **Bash** (Linux-based data handling and pipeline execution).

Research skills: Experimental design and optimization, data interpretation and statistical analysis, scientific writing (manuscripts, theses, proposals), research presentation and seminar delivery.

Teaching and academic instruction: Undergraduate lecturing, laboratory module design, assessment preparation and examination grading, thesis student mentorship, curriculum review and development.

Awards and Fellowship

- **National Science and Technology (NST) Fellowship - 2024**
- **Provost Award - 2023, Amar Ekushey Hall, University of Dhaka**
- **Provost Award - 2022, Amar Ekushey Hall, University of Dhaka**
- **Dean's Award - 2022, Faculty of Biological Sciences, University of Dhaka**
- **Dhaka University Honors Scholarship - 2022**

References

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Mohammad Mamun Alam, PhD
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